

Technology #07

Self-Powered Sensing via Piezoelectric Energy
Harvesting

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Self-Powered Sensing via Piezoelectric Energy Harvesting

WHY DOES IT MATTER?

Chronic ICP monitoring requires battery or percutaneous leads

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 5 governing equations, 11 design inputs, 7 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Piezoelectric Energy Harvesting + Implantable Sensing. 6-step build plan with specific test articles and acceptance criteria. Target buyer: Implantable sensor company; neuroshunt manufacturer.

WHAT DOES THE BUYER GET?

Self-powered sensing concept + R1 correction (duty-cycled architecture); Piezo harvesting governing equations + power budget model; Bench test protocols (power, fatigue, biocompatibility); External precedent catalog (PVDF, PZT, lead-free piezo, harvesting ICs); Critical UNKNOWN disclosure (strain amplitude, chronic depoling, biocompatibility of piezo materials); This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

Production piezo transducer design (material, geometry, integration with catheter); Power conditioning circuit (production PCB, miniaturized); Sensor selection + integration + self-referencing architecture; Manufacturing process for catheter-integrated piezo + circuit + sensor; Regulatory submission (active implantable, ISO 14708-1, IEC 60601-1-2); Clinical validation (accuracy + longevity); Long-term reliability testing (fatigue, depoling, biocompatibility)

WHAT IS THE NEXT DECISIVE EXPERIMENT?

Piezo transducer (PVDF or PZT) on shaker table

WHAT WOULD MAKE US KILL IT?

Harvested power at in-vivo strain levels is insufficient for any duty cycle (model shows <0.1 uW average)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 6-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: Piezo transducer (PVDF or PZT) on shaker table

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - Piezo transducer (PVDF or PZT) on shaker table - 8 weeks

Basis: Build plan WP. Scope: V_out and P_out vs strain amplitude and frequency. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Integrated piezo + circuit + sensor - 10 weeks

Basis: Build plan WP. Scope: Duty-cycled sensor operation. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Biocompatibility specimens (PVDF piezo film + parylene-C encapsulation) - 12 weeks (external lab)

Basis: Build plan WP. Scope: ISO 10993 series. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 5 equations. Key assumptions: CSF pulsation provides reliable 1 Hz strain source (literature-supported for cardiac-driven ICP waveform); PVDF or PZT retains piezoelectric properties in implant environment (UNKNOWN for chronic); Rectification efficiency 50-80% achievable at low voltage (EXTERNAL_PRECEDENT); Energy storage leakage negligible over measurement cycle (UNKNOWN for implantable capacitors); Sensor active power target $\sim 10 \mu\text{W}$ (EXTERNAL_PRECEDENT for low-power medical sensors). UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: Piezo transducer (PVDF or PZT) on shaker table. Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Piezoelectric transducer harvests energy from CSF pulsation; duty-cycled sensing matches harvested power budget (R1 fix)

POTENTIAL VALUE DRIVER: Addresses: Chronic ICP monitoring requires battery or percutaneous leads

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Implantable sensor company; neuroshunt manufacturer

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Piezoelectric Energy Harvesting + Implantable Sensing

Disciplines: Vibration mechanics, Piezoelectric transducer engineering, Energy harvesting circuit design, Fatigue and reliability engineering, Embedded sensing

Architecture: Self-powered CSF pressure/flow sensor powered by harvested mechanical energy from CSF pulsations or catheter motion. R1 fix: original P-15 assumed unrealistic power density; R1 redesign uses piezoelectric transducer with rectifier + storage + duty-cycled sensing.

Subsystems:

SS-01: Piezoelectric transducer (PVDF or PZT) - Converts mechanical strain (CSF pulsation, catheter motion) to electrical charge via piezoelectric effect

SS-02: Power conditioning (rectifier + boost + storage) - AC->DC rectification, voltage boost, energy storage in capacitor or thin-film battery

SS-03: Duty-cycled sensor (pressure or flow) - Measures ICP or flow at low duty cycle (e.g., 1 measurement per minute) compatible with harvested energy budget

SS-04: Wireless data transmitter (passive RFID or backscatter) - Transmits measurement to external reader without requiring onboard power for active RF

2. Mechanism Architecture

Physical Changes: Mechanical strain deforms piezoelectric transducer, generating charge $Q = d_{33} * F$ (force) or $V = g_{33} * d * \sigma$ (stress); rectified and stored; sensor sampled at duty cycle

Key Physics: Piezoelectric effect in PVDF or PZT: charge generated proportional to applied stress. Power output $P \sim (d_{33} * g_{33} * \sigma^2 * \text{Volume}) / (2 * \text{cycle_time})$, scales with stress squared and active volume.

3. Governing Engineering Model

Summary: Piezoelectric energy harvesting from low-frequency mechanical strain. Power output scales with stress squared, active volume, and material coupling coefficient. Duty-cycled sensing matches available power.

Equations:

$V = g_{33} * d * \sigma$ [piezoelectric voltage]

$P \sim (d_{33} * g_{33} * \sigma^2 * \text{Volume}) / (2 * \text{duty_cycle})$ [piezo harvested power]

$P_{\text{avg}} = (d_{33} * g_{33} * \sigma_{\text{rms}}^2 * \text{Volume} * \eta) / (2 * T_{\text{cycle}})$

$E_{\text{stored}} = 0.5 * C * V^2$ [capacitor storage]

$f_{\text{duty}} = P_{\text{harvested}} / P_{\text{sensor_active}}$ [achievable duty cycle]

Assumptions:

- CSF pulsation provides reliable 1 Hz strain source (literature-supported for cardiac-driven ICP waveform)
- PVDF or PZT retains piezoelectric properties in implant environment (UNKNOWN for chronic)
- Rectification efficiency 50-80% achievable at low voltage (EXTERNAL_PRECEDENT)
- Energy storage leakage negligible over measurement cycle (UNKNOWN for implantable capacitors)
- Sensor active power target $\sim 10 \mu\text{W}$ (EXTERNAL_PRECEDENT for low-power medical sensors)

Boundary Conditions:

- Strain amplitude: $\sim 10\text{-}100 \mu\epsilon$ typical for physiologic motion (EXTERNAL_PRECEDENT)
- Frequency: 1 Hz (cardiac) + postural transients
- Operating temperature: 37°C constant

Failure Regimes:

- Stress below threshold $\rightarrow$ harvested power $<$ sensor standby $\rightarrow$ system never wakes
- Piezo depoling at body temperature or due to mechanical fatigue $\rightarrow$ d_{33} degrades
- Capacitor leakage $>$ harvested energy $\rightarrow$ no net storage
- Rectifier forward voltage drop $>$ piezo output voltage $\rightarrow$ no rectification

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Self-powered sensing enables chronic monitoring without battery replacement or periodic leads (R332)	ENGINEERING_PROPOSED
DI-002	Functional requirement	Self-powered pressure/flow sensing via piezoelectric energy harvesting	MODELLED
DI-003	Mechanical strain source	CSF pulsation (~1 Hz, cardiac-driven) + postural motion (R332 + external literature)	EXTERNAL_PRECEDENT
DI-004	Piezo material constants	EXTERNAL_PRECEDENT (PVDF d33=-33 pC/N, g33=0.216 V·m/N; PZT-5B d33=593 pC/N, g33=0.019 V·m/N)	EXTERNAL_PRECEDENT
DI-005	Sensor power target	EXTERNAL_PRECEDENT (~10 µW active, ~0.1 µW standby for low-power external use)	EXTERNAL_PRECEDENT
DI-006	Duty cycle target	MODELLED ~1% based on power budget (R1 fix)	MODELLED
DI-007	Biocompatibility (ISO 10993)	UNKNOWN — especially PZT with lead	UNKNOWN
DI-008	Sterilization	UNKNOWN — piezo may be gamma-sensitive; aseptic or EtO	UNKNOWN
DI-009	EMC	UNKNOWN — backscatter transmission low power but EMC required	UNKNOWN
DI-010	Active implantable standard	UNKNOWN — likely ISO 14708-1 applies	ENGINEERING_PROPOSED
DI-011	Sensor drift target	UNKNOWN — R1 fix uses self-referencing architecture	MODELLED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Piezo transducer geometry + material selection	ABSENT	Final material (PVDF vs PZT vs lead-free); Volume target; Geometry into
DO-002	Power conditioning circuit	CONCEPTUAL	
DO-003	Sensor subsystem (pressure or flow)	CONCEPTUAL	
DO-004	Backscatter transmitter	CONCEPTUAL	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Piezo material d33	EXTERNAL_PRECEDENT (PVD/PC#33 pC/N; PZT#593 pC/N)	pC/N	Material constants	
Piezo material g33	EXTERNAL_PRECEDENT (PVD/PC#16 V·m/N; PZT#593 V·m/N)	V·m/N	Material constants	
Stress amplitude sigma	UNKNOWN for CSF environment (EXTERNAL_PRECEDENT for similar implants: 10-100 MPa)	MPa	Precedent for similar implants	In vitro strain measurement
Active piezo volume	UNKNOWN	mm^3	Design choice constrained by catheter geometry	Geometric specification
Rectification efficiency eta	EXTERNAL_PRECEDENT (50-80% at low voltage)	%	Standard rectifier circuit performance	
Sensor active power	EXTERNAL_PRECEDENT (~10 mW for low-power medical sensors)	mW	Power budget / sensor specs	
Duty cycle target	MODELLED (~1% based on power budget)	%	Energy budget calculation	
Piezo fatigue lifetime	UNKNOWN for chronic strain cycles	Cycles	UNKNOWN in vivo	ASTM E466 fatigue test

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Insufficient harvested power	Stress amplitude too low or piezo area too small	Pre-assembly testing + high-resolution material (PZT) selection	Electromechanical modeling	UNKNOWN in vivo
Piezo depoling	Thermal or mechanical stress	Material selection + strain de-rating	Thermal cycling + 37°C field strain cycling	UNKNOWN long-term depoling rate
Mechanical fatigue of piezo	Cyclic strain causes material degradation	Material selection + strain de-rating	ASTM E466 fatigue characterization	UNKNOWN fatigue lifetime at operating strain
Capacitor leakage excursions	Storage capacitor self-discharge / leakage	Large leakage capacitor selection	Leakage measurement	UNKNOWN achievable leakage
Rectifier forward voltage drop	Piezo output voltage < rectifier forward voltage	Active rectifier or voltage multiplier	Circuit simulation + bench	UNKNOWN minimum piezo voltage
Biocompatibility failure	Piezo material (especially PZT with Pb)	Use of Pb-free materials (e.g. PZT-free piezoelectric series)	ISO 10993 biocompatibility testing	UNKNOWN chronic biocompatibility of piezo materials
Sensor drift	Long-term sensor offset drift	Self-referencing architecture (ring buffer) + calibration	Long-term stability testing	UNKNOWN achievable drift

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Insufficient harvested power	Stress amplitude or piezo volume too small	Increase volume or use higher-coupling material	UNKNOWN actual sensor power
Piezo depoling	Thermal/mechanical stress exceeds Curie temp	Material selection + derating	UNKNOWN chronic depoling rate
Mechanical fatigue of piezo	Cyclic strain crack propagation	Material selection + strain derating	UNKNOWN fatigue lifetime
Capacitor leakage exceeds self-charge	Self-charge faster than recharge	Low-leakage capacitor + shorter cycle	UNKNOWN achievable leakage
Rectifier forward voltage too low	Forward voltage < rectifier drop	Active rectifier or voltage multiplier	UNKNOWN minimum piezo voltage
Biocompatibility failure	Piezo material not biocompatible (esp. PZT)	PZT preferred; hermetic encapsulation	UNKNOWN chronic biocompatibility
Sensor drift	Long-term offset drift between measurements	Self-referencing architecture (R1 fix) + calibration	UNKNOWN achievable drift

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Harvested power meets duty cycle target	Bench with representative strain	$P_{avg} * \eta \geq P_{sensor} * f_{duty}$	NOT_TESTED
V-002	Piezo fatigue lifetime > 10^9 cycles	ASTM E466	Lifetime > 10^9 cycles	NOT_TESTED
V-003	Biocompatibility ISO 10993	ISO 10993 series	Pass	NOT_TESTED
V-004	Sensor accuracy over duty cycle	Calibration + drift	Within spec between calibrations	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Continuous in vivo operation	Animal implant study	Operates target duration without re-power	NOT_PERFORMED
VAL-002	Clinical measurement accuracy	Clinical study vs reference ICP	Clinically acceptable	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Piezo transducer	PVDF (polyvinylidene fluoride) film	Published PVDF harvester literature	ISO 10993 + fatigue + power output	ENGINEERING_CANDIDATE
Piezo transducer	PZT-5H ceramic	Standard PZT material; higher coupling than encapsulated	+ ISO 10993 + fatigue	CANDIDATE — encapsulation required
Piezo transducer	Lead-free piezo (e.g., BaTiO3, KNN)	Published lead-free piezo literature	ISO 10993 + fatigue + power output	CANDIDATE — lower coupling than PZT

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Piezoelectric transducer (PVDF film or PZT chip)	1	CUSTOM_COMPONENT	PVDF or PZT	Multiple (e.g., PI, Murata)	MEDIUM	Calibration + system compatibility + power output
02	Rectifier + boost IC	1	COTS_COMPONENT	SiC (COTS)	Multiple (ADI, TI)	HIGH	Circuit simulation + bench
03	Storage capacitor (low-leakage)	1	COTS_COMPONENT	Tantalum or ceramic	Multiple	HIGH	Leakage measurement at 37°C
04	Sensor (pressure or flow)	1	COTS_COMPONENT	SiC (COTS or custom MEMS)	Multiple (e.g., Sensortek)	MEDIUM	Calibration + drift
05	Backscatter transmitter	1	CUSTOM_COMPONENT	PCB + antenna	Multiple	MEDIUM	Read range test

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no integrated catheter-piezo-sensor process validated

Candidate Processes:

- PVDF film deposition and patterning: Standard PVDF film manufacturing - Requires poling step to activate piezo properties
- Hybrid circuit assembly (rectifier + storage + sensor + transmitter): Standard PCB assembly + chip-on-board - Miniaturization required for catheter integration
- Hermetic encapsulation (if PZT used): Standard implantable device encapsulation (e.g., parylene-C, titanium canister) - Tradeoff between encapsulation robustness and strain transfer

14. External Evidence

Source 1: Conceptual Piezoelectric-Based Energy Harvester from In

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11434573>

Excerpt: by M Khazaei · 2024 · Cited by 11 — This paper studies the development of piezoelectric energy harvesting for self-powered leadless intracardiac pacemakers .

Source 2: Piezoelectric Energy Harvesting for Pacemaker Applications

URL: <https://pubs.acs.org/abseba/article/11/8/4570/3599553/Piezoelectric-Energy-Harvesting-for-Pacemaker>

Excerpt: by D Singh · 2025 · Cited by 13 — This device design produced a mean power of 0.78 μ W for a heart rate of 84 bpm when implanted in the pig's heart at the right apical ventricle.

Source 3: Recent advancements in piezoelectric energy harvesting

URL: <https://journals.sagepub.com/doi/10.1177/1045389X231200144?int.sj-full-text.similar-articles.3>

Excerpt: Oct 21, 2023 — This work focuses on EH for biomedical devices such as pacemakers, gastric stimulators, implantable cardioverter-defibrillator (ICD), etc. A ... Read more

Source 4: Implant harvests heartbeat power | Research

URL: <https://www.chemistryworld.com/news/implant-harvests-heartbeat-power/7002.article>

Excerpt: Jan 21, 2014 — A flexible piezoelectric implant that harnesses energy from the body's natural motions has been developed by researchers in the US and China. Read more

Source 5: Cardiac energy harvesting and sensing based on

URL: <https://www.sciencedirect.com/science/article/am/pii/S2211285520306534>

Excerpt: by L Dong · 2020 · Cited by 127 — The essential energy harvesting mechanism for those reported devices is to harness cardiac motion or pacemaker lead motion into strain within the piezoelectric ...

15. Transfer Boundary

YOU RECEIVE:

- + Self-powered sensing concept + R1 correction (duty-cycled architecture)
- + Piezo harvesting governing equations + power budget model
- + Bench test protocols (power, fatigue, biocompatibility)
- + External precedent catalog (PVDF, PZT, lead-free piezo, harvesting ICs)
- + Critical UNKNOWN disclosure (strain amplitude, chronic depoling, biocompatibility of piezo materials)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > Production piezo transducer design (material, geometry, integration with catheter)
- > Power conditioning circuit (production PCB, miniaturized)
- > Sensor selection + integration + self-referencing architecture
- > Manufacturing process for catheter-integrated piezo + circuit + sensor
- > Regulatory submission (active implantable, ISO 14708-1, IEC 60601-1-2)
- > Clinical validation (accuracy + longevity)
- > Long-term reliability testing (fatigue, depoling, biocompatibility)

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Actual mechanical strain amplitude in CSF/catheter environment — UNKNOWN (critical for power budget)
- Long-term piezo depoling rate in vivo — UNKNOWN
- Achievable duty cycle given real strain — UNKNOWN
- Biocompatibility of piezo materials (especially PZT with lead) in CSF — UNKNOWN
- Sensor drift in duty-cycled operation — UNKNOWN
- Regulatory pathway (active implantable medical device, ISO 14708-1) — UNKNOWN specifics
- Manufacturing integration of piezo + circuitry + sensor in catheter — UNKNOWN

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	Piezo transducer (PVDF)	Shaker 2000, oscilloscope, digital multimeter	Strain amplitude, duty cycle, power output	Strain amplitude > 100 µm/m, duty cycle > 50%, power output > 100 µW	Strain amplitude, duty cycle, power output	12 weeks (chronic)
WP-02	Rectifier + storage circuit	Bench power supply, multimeter, capacitor, inductor	Rectification efficiency, leakage current	Rectification efficiency > 80%, leakage current < 100 nA	Rectification efficiency, leakage current	12 weeks (chronic)
WP-03	Integrated piezo + circuit	Shaker 2000, oscilloscope, digital multimeter	Duty-cycled sensor operation	Sensor produces measurable signal at duty cycle > 50%	Sensor produces measurable signal at duty cycle > 50%	12 weeks (chronic)
WP-04	Piezo fatigue specimen	Fatigue test frame (ASTM E466)	Operating strain	Strain > 100 µm/m	Fatigue report	12 weeks (chronic)
WP-05	Biocompatibility specimen	ISO 10993 Biocompatibility test chamber	ISO 10993 pass	ISO 10993 pass	ISO 10993 report	12 weeks (external lab)

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-06	In vivo strain measure	Animal facility, strain gage amplification	Strain gage amplitude	Strain to detect	Strain to detect	16 weeks

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 5 governing equations and a 6-step build plan. The next experiment is clearly specified. Target buyer: Implantable sensor company; neuroshunt manufacturer.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Harvested power at in-vivo strain levels is insufficient for any duty cycle (model shows <0.1 uW average)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Harvested power at in-vivo strain levels is insufficient for any duty cycle (model shows <0.1 uW average)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.